Here $a \equiv b \pmod{n}$ means $a - b = n \cdot k$ for some integer k and b is the **least non-negative integer** when a is divided by n.

For instance, $25 \equiv 4 \pmod{7}, -20 \equiv -2 \pmod{3} \equiv 1 \pmod{3}$ and $15 \equiv 0 \pmod{5}$, etc. Further the set of integers when divided by n, leaves the remainder $0, 1, 2, \ldots, n-1$. In the case of $\mathbb{Z}_5$,

$$
\begin{aligned}
[0] &= \{\ldots, -15, -10, -5, 0, 5, 10, 15, \ldots\} \\
[1] &= \{\ldots, -14, -9, -4, 1, 6, 11, \ldots\} \\
[2] &= \{\ldots, -13, -8, -3, 2, 7, 12, \ldots] \\
[3] &= \{\ldots, -12, -7, -2, 3, 8, 13, \ldots\} \\
[4] &= \{\ldots, -11, -6, -1, 4, 9, 14, \ldots\}.
\end{aligned}
$$

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We write this as $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$. In each class, any two numbers are congruent modulo 5.

Before 2007, modular arithmetic is used in 10-digit ISBN (International Standard Book Number) numbering system. For instance, the last digit is for parity check. It is from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, X\}$. In ISBN number, 81-7808-755-**3**, the last digit **3** is obtained as

$1*8+2*1+3*7+4*8+5*0+6*8+7*7+8*5+9*5=8+2+21+32+0+48+49+40+45=245 \equiv 3 \pmod{11}$.

Alternatively, the weighted sum is calculated in the reverse manner

9*8+8*1+7*7+6*8+5*0+4*8+3*7+2*5+1*5=245 = **3** (mod 11).

In both ways, we get the same check number 3.

After 2007, 13-digit ISBN numbering has been followed. The first 12 digits (from left to right) are multiplied by the weights 3,1,3,1,…. starting from right to left. Then the weighted sum is calculated. The higher multiple of 10 is taken. Then the difference is calculated. Then its additive inverse modulo 10 is the thirteenth digit.

For instance, consider the ISBN Number: 978-81-931995-6-**5.**Take 12 digits from left to right.

9	7	8	8	1	9	3	1	9	9	5	6
1	3	1	3	1	3	1	3	1	3	1	3
9	21	8	24	1	27	3	3	9	27	5	18

The total of last row is 155. The nearest (higher) integer in multiples of 10 is 160. The difference 160-155=5. The additive inverse modulo 10 is **5** which is 13-th digit in the ISBN number.

Two new operations namely **addition modulo** $n(+_n)$ and **multiplication modulo** $n(\times_n)$ are defined on the set $\mathbb{Z}_n$ of all non-negative integers less than n under modulo arithmetic.

Definition 12.6

(i) The addition modulo n is defined as follows.

Let $a, b \in \mathbb{Z}_n$. Then

$a +_n b$ = the remainder of $a + b$ on division by n.

(ii) The multiplication modulo n is defined as follows.

Let $a, b \in \mathbb{Z}_n$. Then

$a \times_n b$ the remainder of $a \times b$ on division by n

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